

Clinic Patient Visit Monitoring System

ASP.NET Core MVC Laboratory Exercise Specification

Scenario

A community clinic needs a web application for reception staff to register, log in, record patient visits, monitor consultation progress, and mark consultations as completed. No billing, prescriptions, or EMR features are required.

Learning Objectives

ASP.NET Core MVC, Model Binding, Razor Views, DTOs, Data Annotation Validation, Cookie Authentication, Authorization, Repository Pattern, CRUD, Search, Bootstrap.

Functional Requirements

Implement User Registration, Login, Logout, Register Patient Visit, View Patient Monitoring List, Edit Patient Visit, View Details, Complete Consultation, and Search.

Patient Visit Fields

Visit Number (auto-generated), First Name, Last Name, Age, Sex, Contact Number, Address, Physician, Visit Type, Arrival Date & Time, Chief Complaint, Notes.

Monitoring

Display Visit Number, Patient, Physician, Visit Type, Arrival Time, Consultation End Time, Status, and Actions.

Status

Waiting, In Consultation, Completed.

Validation

Use Required, Range, Phone, Display, DataType, and other Data Annotation attributes. Validate ModelState.

Authentication

Public: Login, Register. Protected: Patient pages using Cookie Authentication and [Authorize].

Entities

User: Id, FirstName, LastName, Email, Username, Password. PatientVisit: Id, VisitNumber, FirstName, LastName, Age, Sex, ContactNumber, Address, Physician, VisitType, ArrivalDateTime, ConsultationCompletedDateTime, Status, ChiefComplaint, Notes.

Repositories

UserRepository and PatientVisitRepository using static List.

Controllers

HomeController, AccountController, PatientVisitController.

Views

Login, Register, Index, Create, Edit, Details, Complete.

Submission

Submit a complete ASP.NET Core MVC solution with authentication, validation, repository pattern, CRUD, search, Bootstrap UI, and clean code.